SIMI KUMARI

AI/ML | GENERATIVE AI | NLP | BACKEND

+91-9101573795 | simikumari571@gmail.com | GitHub | LinkedIn | Portfolio

Education

IKG Punjab Technical University, Punjab, India — **Jul 2023 – Jun 2027**
B.Tech in Computer Science and Engineering — **CGPA: 8.6/10**

Experience

FlyRank AI — **Jul 2026 – Aug 2026**
AI Backend Engineer Intern — *Remote*

- Built and shipped **RESTful AI backend services** with Python and FastAPI that connected frontend features to AI models and databases, enabling production data-driven workflows.
- Owned end-to-end integration across **API, database, and AI layers**—resolved critical integration issues and delivered reliable, production-oriented backend components.

Projects

BeliefGraph – Dependency-Aware Recovery for AI Agents — GitHub | Live Demo
Python, FastAPI, Pydantic, PostgreSQL, Docker, Pytest, React, TypeScript

- Designed a dependency graph modeling relationships between beliefs, conclusions, plans, and actions to analyze downstream impact when upstream assumptions change or actions fail.
- Implemented dependency-aware selective recovery to invalidate and recompute affected nodes while preserving valid branches, reducing derived-node recomputation by **57%** versus full recomputation while maintaining **100% correctness in controlled benchmarks**.
- Developed a FastAPI backend with Pydantic models, Docker packaging, and automated Pytest coverage; exposed recovery states through REST APIs and a React/TypeScript frontend.
- Integrated LangChain and Groq into LLM-backed agent workflows to demonstrate evidence-informed recovery decisions without restarting the entire workflow.

MedIntel AI – Healthcare Intelligence Platform — GitHub | Live Demo
Python, FastAPI, React, TypeScript, Tailwind CSS, PostgreSQL, LangChain, Groq API, Tesseract OCR, PyMuPDF

- Built a **full-stack AI healthcare platform** supporting medical report interpretation, medication guidance, drug interaction analysis, BMI calculation, and medical image analysis.
- Designed a **RAG pipeline** using LangChain, vector embeddings, and LLMs to generate context-aware responses from uploaded medical documents.
- Developed a **document-processing pipeline** using Tesseract OCR and PyMuPDF to extract information from scanned and text-based reports.
- Implemented FastAPI REST services, database persistence, and conversation history to support end-to-end application workflows.

AI Learning Companion – RAG Conversational AI — GitHub | Live Demo
Python, LangChain, ChromaDB, Sentence Transformers, Groq API, Streamlit

- Engineered an end-to-end **RAG question-answering system** covering PDF ingestion, document chunking, embedding generation, semantic retrieval, and LLM response generation.
- Indexed **100+ PDF knowledge sources** using Sentence Transformers and ChromaDB to support context-aware document retrieval.
- Implemented vector similarity search and integrated Groq LLM inference to generate responses grounded in retrieved document content.

Technical Skills

Programming: Python, SQL, JavaScript, TypeScript, Data Structures & Algorithms
AI/ML: Machine Learning, Deep Learning, NLP, Transformers, BERT, Model Fine-Tuning, Feature Engineering, Model Evaluation
Generative AI & Agents: LLMs, RAG, Prompt Engineering, LangChain, Hugging Face Transformers, Sentence Transformers, Embeddings, Semantic Search, Dependency Graphs for Agent Recovery
Backend & AI Systems: FastAPI, REST APIs, Pydantic, Flask, API Integration, Backend Development
Databases & Retrieval: PostgreSQL, SQLite, ChromaDB, Vector Search
Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy
Tools & Deployment: Git, GitHub, Docker, Pytest, Streamlit, Render, Hugging Face Spaces, Vercel, React

Certifications

Generative AI with LangChain and Hugging Face – Udemy — **Jun 2026**
Data Science and Machine Learning – Independent Learning — **Dec 2025**
Data Science with AI – Internshala Trainings — **Jul 2025**